

WEEK 8 – POWER BI STUDENT PERFORMANCE ANALYSIS REPORT

1. Introduction:

The **Student Performance Analysis Dashboard** is developed using **Microsoft Power BI** to analyse and present academic performance data in a structured and visually effective manner. The dashboard provides an overview of student performance in **Mathematics, Reading, and Writing**, along with the overall average score and pass percentage.

The analysis further examines student performance across different categories such as **race/ethnicity, parental level of education, lunch category, test preparation course, and gender**. Various charts, performance indicators, and interactive filters are incorporated to make the information easier to interpret and compare.

The primary aim of the dashboard is to transform student performance data into **clear and meaningful visual insights**. Power BI helps present the data in an organized format, making it easier to identify performance patterns, compare different student groups, and understand the overall academic performance.

2. Objectives:

The major objectives of the Week 8 dashboard are:

- To analyse student performance using Microsoft Power BI.
- To calculate and display average scores for different subjects.
- To understand overall student performance through summary indicators.
- To compare student scores among different race/ethnicity groups.
- To study the relationship between student performance and parental education level.

- To compare academic performance based on lunch category.
- To analyse student performance according to test preparation course status.
- To compare the performance of male and female students.
- To use interactive filtering for analysing selected student groups.
- To present the student dataset through simple and meaningful visualizations.

3. Dashboard Overview:

The Student Performance Dashboard provides a visual summary of the academic performance data. It includes **average score indicators, comparison charts, category-based visualizations, and interactive filtering options.**

The dashboard presents subject-wise performance in Mathematics, Reading, and Writing and also calculates an overall average score. Other visuals allow the performance to be examined according to different student characteristics.

4. Dashboard Components:

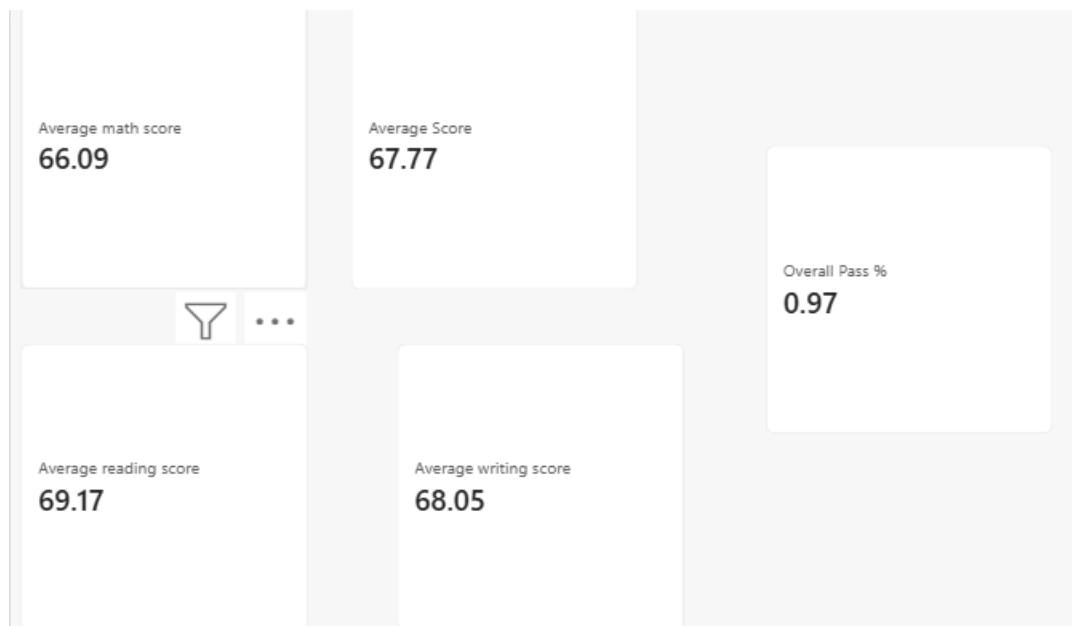
Average Performance Indicators

The dashboard displays important average performance values for the three subjects and the overall score.

The values shown in the dashboard are:

- **Average Math Score - 66.09**
- **Average Reading Score - 69.17**
- **Average Writing Score - 68.05**
- **Average Overall Score - 67.77**

These indicators provide a quick summary of the students' academic performance. Among the three subjects, Reading has the highest average score, followed by Writing and Mathematics.



Overall Pass Percentage

The dashboard also includes an **Overall Pass %** indicator. The value displayed in the main dashboard is **0.97**.

This indicator gives a summarized view of the overall passing performance in the dataset. It allows the user to understand the pass-related performance quickly without going through individual student records.

Average Scores by Lunch Category

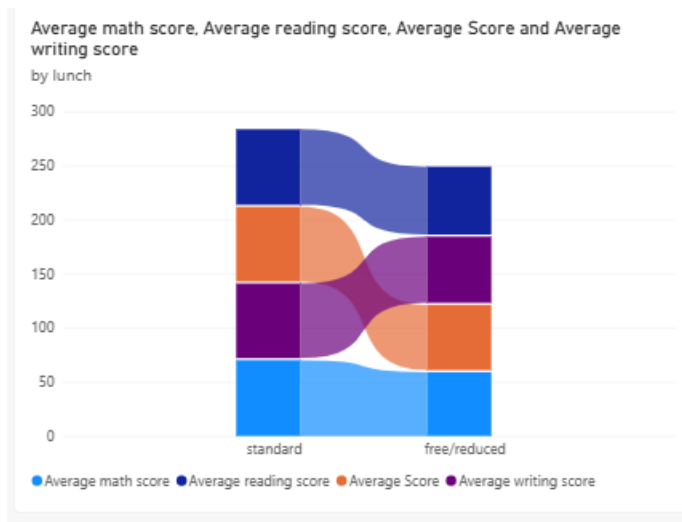
The dashboard compares student performance according to two lunch categories:

- **Standard**
- **Free/Reduced**

The visual includes Mathematics, Reading, Writing, and Overall Average Score. This allows the performance of the two groups to be compared using multiple measures.

For example, the dashboard shows the average Mathematics score as **70.03 for Standard lunch** and **58.92 for Free/Reduced lunch**.

This visualization provides a clear comparison of student performance between the two lunch categories



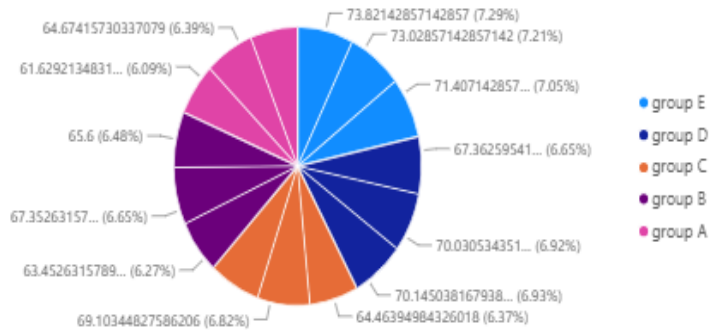
Performance by Race/Ethnicity

The dashboard contains a visual comparing the average subject scores among **Group A, Group B, Group C, Group D, and Group E**.

The visualization represents Mathematics, Reading, and Writing performance for the different groups. It helps identify variations in academic scores across the five race/ethnicity categories.

The visual provides a category-wise comparison and makes it easier to identify which groups have relatively higher or lower average performance.

Average math score, Average reading score and Average writing score
by race/ethnicity



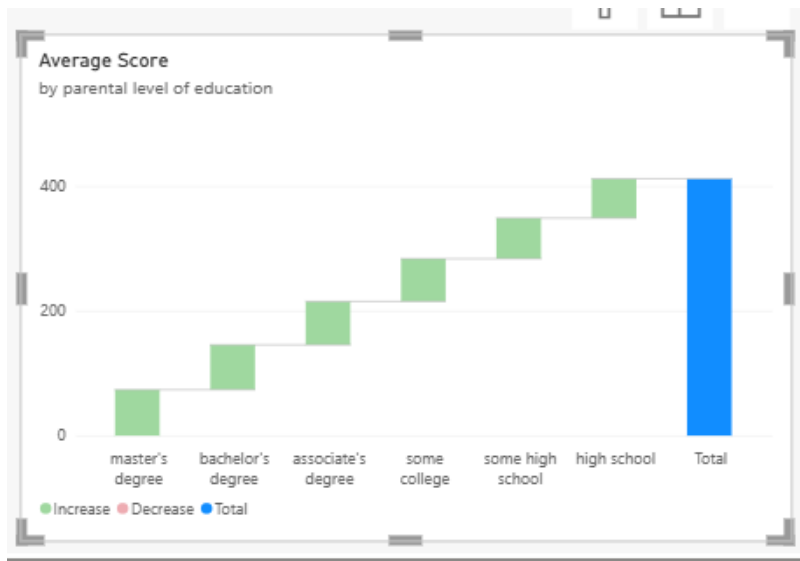
Average Score by Parental Level of Education

Another visualization in the dashboard analyses the **Average Score according to parental level of education**.

The categories represented include:

- Master's degree
- Bachelor's degree
- Associate's degree
- Some college
- Some high school
- High school

The chart helps compare the overall academic scores of students across different parental education categories. It provides a visual way to examine how student performance is distributed across the different education levels.



Subject Performance Based on Test Preparation

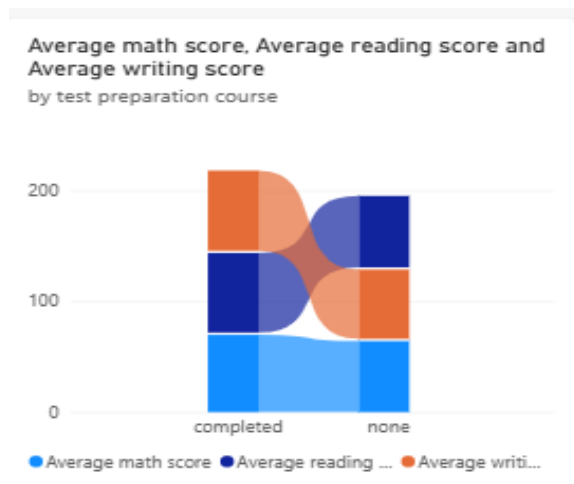
The dashboard also compares student performance according to **test preparation course status**.

The two categories shown are:

- **Completed**
- **None**

The visual compares the average Mathematics, Reading, and Writing scores for these categories. This allows the performance of students who completed the preparation course to be viewed alongside students who did not complete it.

This comparison helps in understanding the differences in displayed subject scores between the two preparation groups.



Performance Comparison by Gender

The dashboard includes a gender-based comparison of student performance.

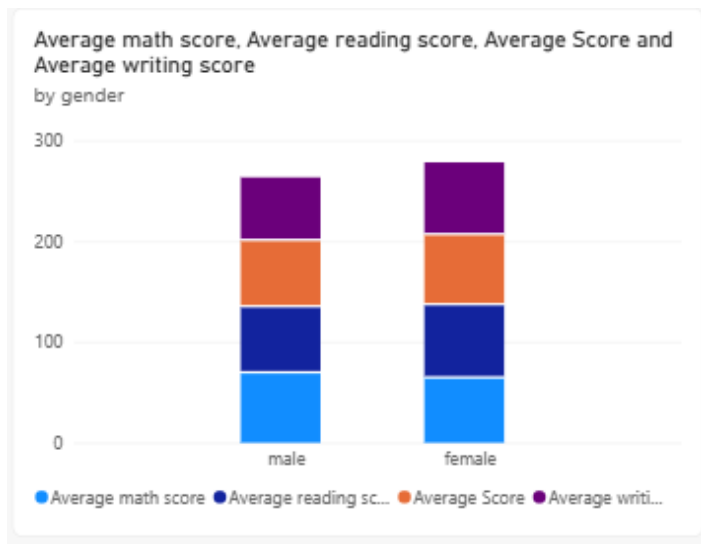
The categories are:

- **Male**
- **Female**

The visual compares:

- Average Math Score
- Average Reading Score
- Average Overall Score
- Average Writing Score

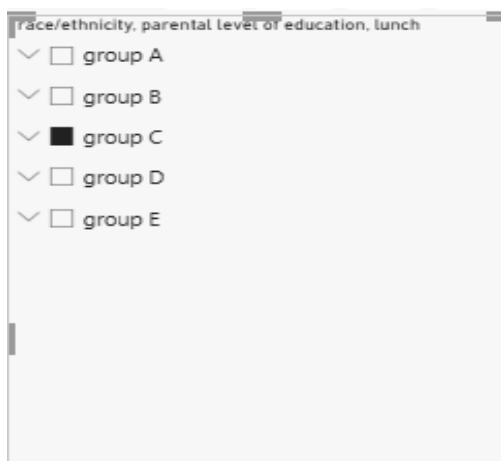
This provides a simple way to observe the differences in average academic performance between male and female students.



Interactive Race/Ethnicity Filter

The dashboard contains a **Race/Ethnicity filter** with Groups A, B, C, D, and E.

This filter allows a particular group to be selected so that the dashboard can be examined based on the chosen category. Interactive filtering makes the dashboard easier to explore and allows the user to focus on specific groups while analysing student performance.



5. Key Observations:

The dashboard provides several useful observations about the student performance dataset.

The **Average Reading Score is 69.17**, which is the highest among the three individual subject averages displayed. The **Average Writing Score is 68.05**, while the **Average Math Score is 66.09**. The overall average score shown in the dashboard is **67.77**.

The lunch-based comparison shows that the displayed average Mathematics score for the Standard category is **70.03**, whereas the Free/Reduced category shows **58.92**.

The dashboard also provides comparisons across race/ethnicity, parental education, test preparation course, and gender. These visuals make it easier to identify differences in student performance between various categories.

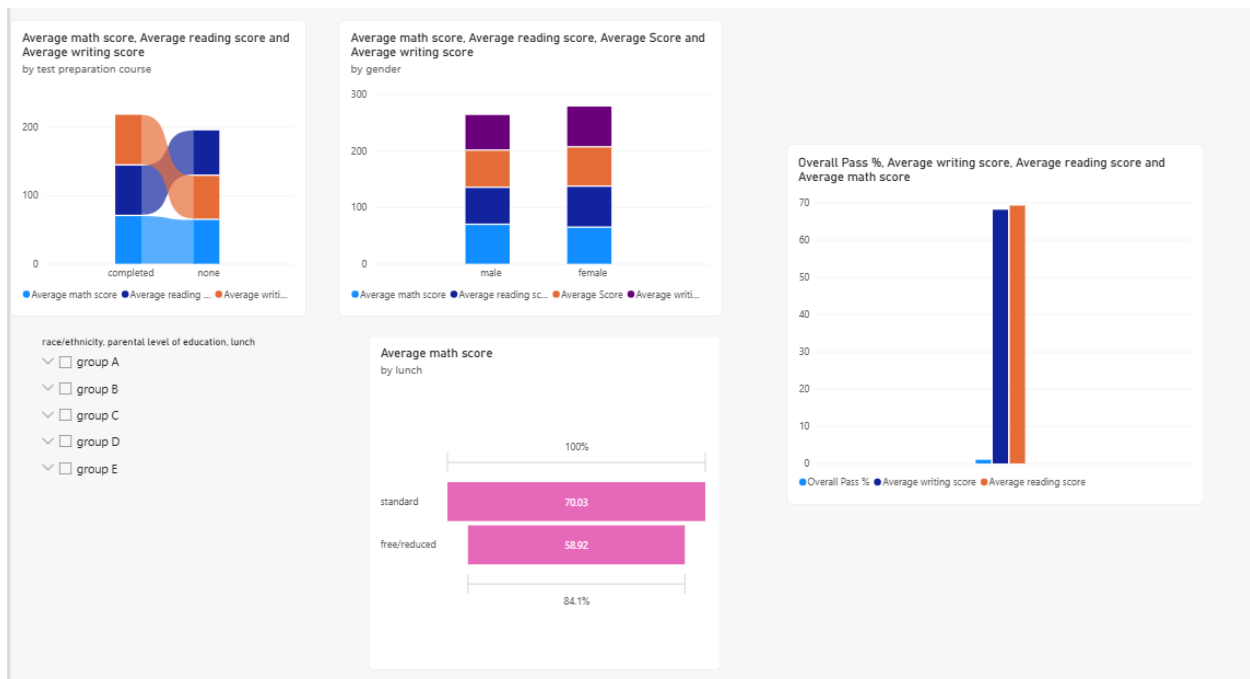
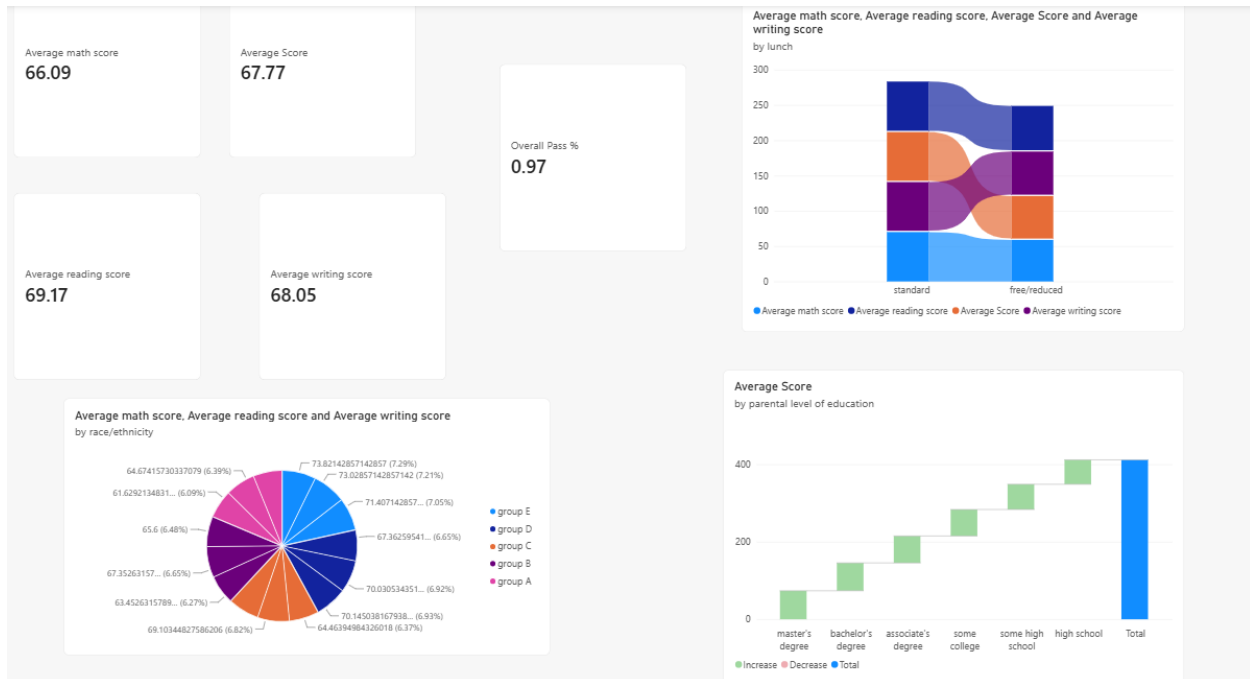
6. Learning Outcomes:

Through this Week 8 activity, I gained practical experience in creating and presenting a **Power BI analytical dashboard**.

I learned how to:

- Create meaningful visualizations from a dataset.
- Display average values using performance indicators.
- Compare multiple categories using charts.
- Analyse student performance from different perspectives.
- Use filters to interact with dashboard data.
- Present analytical results in a simple visual format.
- Understand how Power BI can be used to convert raw data into meaningful information.

This activity improved my understanding of data visualization, dashboard development, data comparison, and interactive analysis using Power BI.



7. Conclusion:

The Week 8 **Student Performance Analysis Dashboard** successfully presents student academic performance through Microsoft Power BI. The dashboard combines subject-wise average scores, overall performance indicators, and different category-based comparisons to provide a complete visual overview of the dataset.

The dashboard analyses student performance based on **lunch category, race/ethnicity, parental level of education, test preparation course, and gender**. The use of interactive filtering also makes it easier to focus on particular student groups.

Overall, this activity helped me understand how Power BI can be used to transform student performance data into clear and meaningful visual information. It also improved my practical knowledge of creating dashboards, comparing data, and presenting analytical findings effectively.

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